6-3

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering **Part-I Odd** Semester Examination **2020**
Course Code: **CSE1111** Course Title: **Introduction to Computer Systems**
Time: **03** Hours Full Marks: **52.5**

[N.B. Answer 06(Six) questions taking any 03(Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1.(a)	Define computer. Explain the characteristics of computer.	3
(b)	What is meant by computer generations? Why were the first generation computers huge in size? State the features of the third generation computers.	3.75
(c)	Distinguish between mainframe and supercomputer.	2
2. (a)	What is CPU? Draw the internal block diagram of a CPU.	3.75
(b)	Explain the importance of a bus in the computer system. What are the different types of buses found in the computer system?	3
(c)	Why cache memory is needed in a computer system? Explain it with block diagram.	2
3. (a)	What is chipset? Explain the main activities of the chipset on a motherboard.	3
(b)	Define computer port. Give the two examples of each of serial port and parallel port.	2
(c)	What is core processor? Explain core i5 processor. Which type of RAM is used in core i5 processor?	3.75
4. (a)	Briefly explain the types of data storage and its relation with CPU.	3
(b)	Explain the principle to store data in magnetic and optical storage devices.	3
(c)	Give a complete specification to buy a modern computer.	2.75

Section-B

5. (a)	Classify various types of Memory on the basis of the material used.	3
(b)	Explain Memory Hierarchy according to access time required for memory.	2
(c)	How many address and data lines will be there for a 16M x 32 memory system?	1.75
(d)	Explain about SSD. Discuss advantages of it.	2
6. (a)	Explain different types of OS and also explain different types of tasks done by OS.	3
(b)	Write three utility programs with its purposes.	2.75
(c)	Explain the DOS internal structure.	3
7. (a)	What is computer network? Give the comparison of different types of computer network.	3.5
(b)	Define topology. Explain the factors that affect choice of topology for a network.	2.25
(c)	Explain the function of following network devices (i) Router, (ii) Switch, (iii) Gateway.	3
8. (a)	Define Malware. Explain how does it spread in computer and how does protect from it.	2
(b)	What is data processing? Explain different stages of data processing cycle.	3
(c)	What is data redundancy and data inconsistency? How DBMS is superior to conventional file processing system?	3.75

Scanned with CamScanner

Dept. Computer Science and Engineering
University of Rajshahi

Semester Final Examination, B.Sc. Engg. 2019, 1st year, Odd semester

Course ID: CSE 1111 Course Title: Introduction to Computer Systems/Computer Fundamentals

Total Time 3 Hours Total Marks 52.5

(Answer any six questions taking three from each section)

Section A

1(a)	What do you mean by system boot up? Briefly discuss the PC's booting up process.	3
(b)	What is meant by computer generations? Explain different generations of computers.	3.75
(c)	Expand ABC, ENIAC, EDVAC and EDSAC.	2
2(a)	What is motherboard? Write the functions of it. Also mention the main components on a motherboard.	3
(b)	Briefly discuss how a keyboard works.	3
(c)	Explain the internal structure of CPU with diagram	2.75
3(a)	How does the control unit assist the CPU in carrying out its operations? Explain.	3
(b)	What do you understand by machine cycle? Explain the main operations those are accomplished during the machine cycle.	3
(c)	How does the bus width affect the overall speed of the computer system? Define the three types of I/O buses.	2.75
4(a)	Define printer. Explain the various criteria to evaluate a printer.	3
(b)	Describe the LCD monitor with its advantages.	3
(c)	Explain the four areas created on a magnetic disk during formatting.	2.75

Section B

5(a)	What are difference between SRAM and DRAM? Why is a refreshing circuit needed for DRAM?	3
(b)	Write the characteristic of cache memory. What do you mean by L1, L2 and L3 cache?	2
(c)	Explain how data is stored on the surface of magnetic and optical disk	3.75
6(a)	Define the software. List and explain the types of software. Give two exaples of each category.	3
(b)	Explain three categories of operating systems with examples.	2.75
(c)	Define utility program. List a few names of utility programs with their tasks.	2
(d)	What is kernel?	1
7(a)	What is a computer network? Illustrate advantages of computer networks.	2.75
(b)	Explain the advantages and disadvantages of client-server and-peer-to-peer networks	3
(c)	Define network topology. Write down the advantages and disadvantages of bus, ring and star topologies.	3
8(a)	Define computer virus. How does computer virus work? Mention some notable computer viruses.	3
(b)	What is data processing? Define data and information.	2
(c)	Define Database Management System (DBMS). Explain the limitations of file based systems and write the advantages of DBMS.	3.75

Scanned with CamScanner

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2018
Course Code: **CSE1111** Course Title: **Introduction to Computer Systems**
Marks: 52.50 Time: 3 Hours

(Answer Six (06) Questions taking any Three (03) Questions from each Section)

Section-A

1.(a)	What is Computer? Explain the characteristics of Computer.	3
(b)	What are the basic operations of Computer? Describe them.	3.75
(c)	Compare microcomputers, minicomputers and main frames in terms of size and cost.	2
2. (a)	What are the different forms to represent negative signed binary numbers? Explain them with example.	3
(b)	Evaluate the binary number 10001111 in base 10 under (i) Signed magnitude notation and (ii) Two's complement notation.	3
(c)	Convert the followings (i) $(510.125)_{10}=()_2$? (ii) $(AEF5)_{16}$ to binary and octal.	2
(d)	Perform the subtraction 11110111-10110111 using 2's complement method.	0.75
3. (a)	Briefly describe the PC's system booting-up sequence.	3
(b)	Describe the function of the CPU as fetching and executing instructions stored in memory.	3
(c)	Explain the different factors affecting the processing speed of CPU.	2.75
4. (a)	Differentiate between LCD and LED monitors.	2.75
(b)	Discuss the printing mechanism of a laser printer.	3
(c)	Define the following terms: (i) Optical Mouse, (ii) Bar-code reader, and (iii) Modem.	3

Section-B

5. (a)	What is Memory? Draw the memory hierarchy according to their access speed.	2
(b)	What are the features of storage devices? Describe them.	3
(c)	Explain how data is stored on the surface of magnetic disk.	2
(d)	A 6 platter hard disk has 600 tracks per surface. There are 10 sectors per track and 512 bytes per sector. What is the storage capacity of the disk? How many cylinders does the disk pack have?	1.75
6. (a)	Define Operating System (OS) and explain functions of OS.	2.5
(b)	Explain three categories of OS with examples.	3
(c)	Differentiate between DOS and Windows.	2
(d)	Why is virtual memory needed?	1.25
7. (a)	Define computer networks. Mention four benefits that computer networks provide to their users.	2.75
(b)	Explain network topology. Discuss the function of Star and Bus topology and differentiate between them.	4
(c)	Define the following terms (i) IP address (ii) TCP/IP, (iii) SMTP, (iv) http.	2
8. (a)	Mention three utility programs with its purposes.	3
(b)	Define computer virus. Explain the characteristics of it.	2.5
(c)	How do viruses infect and spread in a computer system?	2
(d)	Mention different types of viruses with examples.	1.25



Scanned with CamScanner